



External pressure on lithium-metal batteries enables longer life and cheaper production

Researchers at Texas A&M University have made a breakthrough in lithium-ion battery technology, crucial for modern electronics and electric vehicles. Using quantum mechanics, they have developed a method to understand the impact of external pressure on lithium-metal batteries, aiming to enhance durability and efficiency. This research, part of the Battery500 Consortium, seeks to improve electric vehicle battery performance in alignment with US Department of Energy goals. They anticipate a tenfold increase in anode energy density by proposing a shift from conventional graphite anodes to lithium metal. Their study reveals that external pressure can promote uniform lithium-ion distribution on the anode, preventing dendrite formation and potential short-circuiting. This discovery opens the door to developing lithium-metal batteries with longer life spans, increased functionality, and cost-effective production, advancing cleaner, more efficient transportation.

New tech based on 18nm process enhances microcontroller features

STMicroelectronics has introduced a new process technology for designing and building transistors in next-generation embedded processing devices. The technology is based on



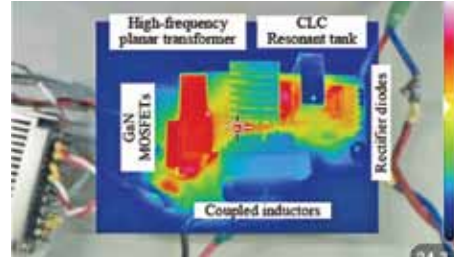
18nm FD-SOI technology with embedded PCM enables next-generation STM32 MCUs (Credit: <https://newsroom.st.com>)

an 18nm fully depleted silicon on insulator (FD-SOI) process with embedded phase change memory (ePCM), reducing the feature size from 20nm.

Developed with Samsung Foundry, this technology improves power consumption, allows for larger memory sizes, and enhances peripheral integration, surpassing the current 40nm eNVM technology. It offers over 50% better performance-to-power ratio, reduces leakage current, minimises interference, and provides higher memory and digital peripheral integration. The technology also enhances RF performance in wireless MCUs, supports 3V operation for analogue features, and ensures reliability for industrial applications. The first microcontroller based on this technology, the STM32, will feature an ARM Cortex-M core optimised for machine learning and digital signal processing, fast memory interfaces, advanced graphics, various peripherals, and advanced security features.

High-frequency boost converter can prove crucial for healthcare and transportation

Kobe University's power electronics team has designed a high-frequency switching boost converter, crucial for energy harvesting and powering medical devices and hydrogen-fuelled cars. This converter efficiently transforms low-voltage



The Kobe University-developed direct current voltage boost converter has much reduced electromagnetic noise and a high energy efficiency of over 91 percent, which is unprecedented for a MHz drive with high voltage multiplier ratio. This ratio is also more than 1.5 times higher than existing designs (Credit: MISHIMA Tomokazu)

age direct current into high-voltage direct current using soft switching techniques. Soft switching ensures zero-voltage transitions, reducing heat loss and increasing efficiency.

The design

incorporates resonant tank circuits for energy storage during switching, minimising losses compared to traditional methods. Using a planar transformer, the team achieved up to 91.3% energy efficiency with reduced electromagnetic noise, surpassing existing designs. This advancement has broad implications for renewable energy, transportation, and healthcare, with plans to enhance the prototype's power capacity in the future.

New approach could better integrate solar and wind energy sources into a grid

Researchers from the Okinawa Institute of Science and Technology and Ben-Gurion University of the Negev have devised a method to anticipate and analyse energy production fluctuations in solar PV systems across various locations. This approach, centred on comprehending the power spectrum, holds significant importance for integrating renewable energy sources like solar and wind into the electricity grid. The research leverages the clear-sky index to gauge solar radiation under ideal conditions, aiding in understanding deviations caused by atmospheric factors such as clouds and aerosols. Unlike wind power, solar power lacks a systematic forecasting method. By analysing data from locations like the Negev desert and an Indian Ocean buoy, the researchers identified changes in solar radiation due to environmental factors and clear-sky patterns. Some energy fluctuations may become less pronounced across different locations, while others may persist.